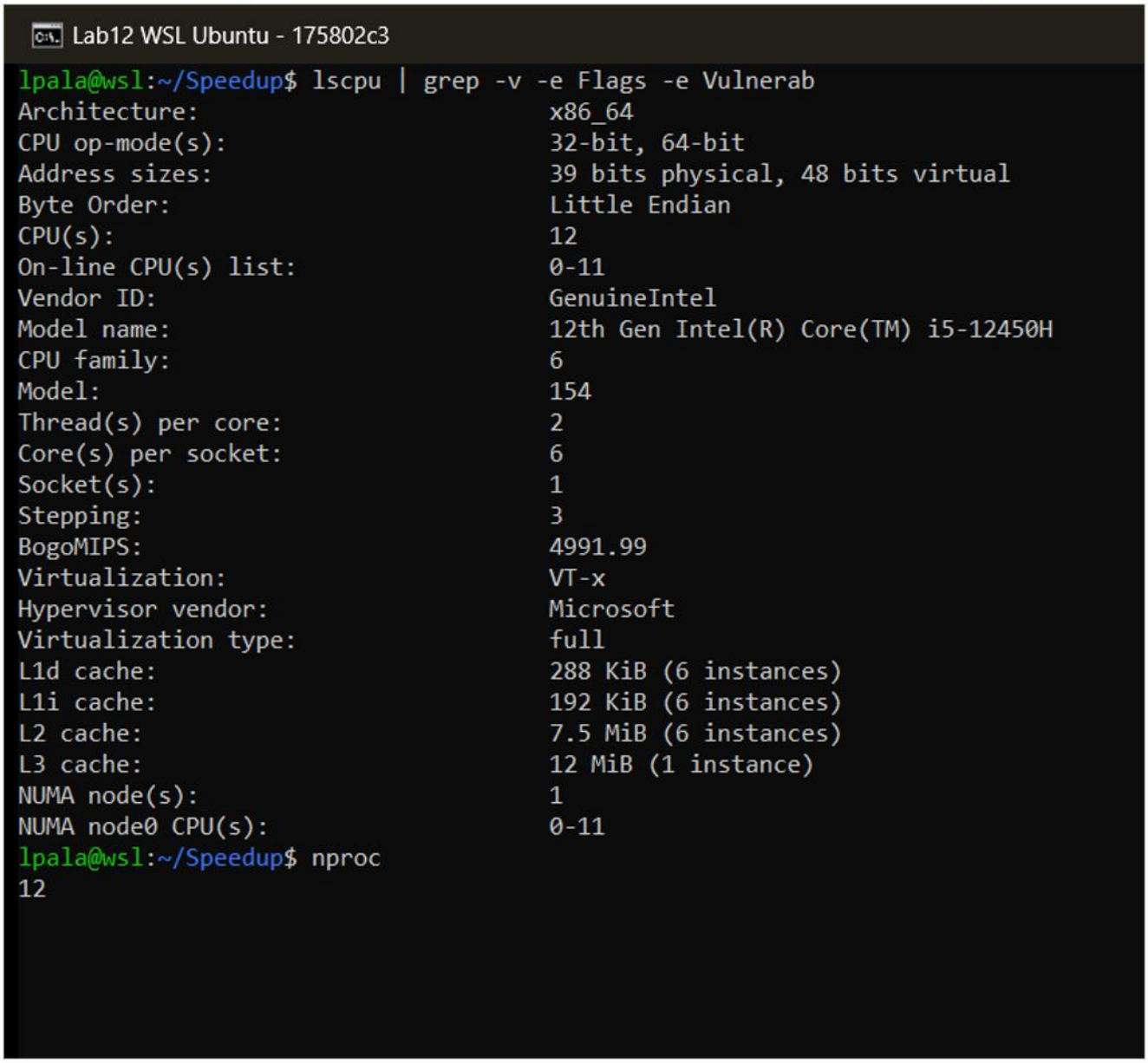
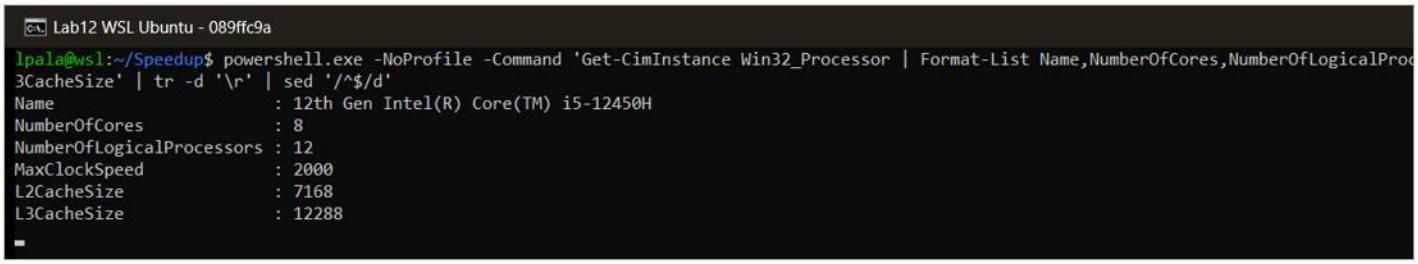
Datos del equipo (WSL2: lscpu)

Dentro de WSL2 lscpu muestra 6 núcleos × 2 hilos: es la vista virtualizada de Hyper-V.



Datos del equipo (Windows)

Vista real: 8 núcleos físicos (4 P con HT + 4 E), 12 hilos lógicos.



Compilación (make)

Lab12 WSL Ubuntu - c1197afa

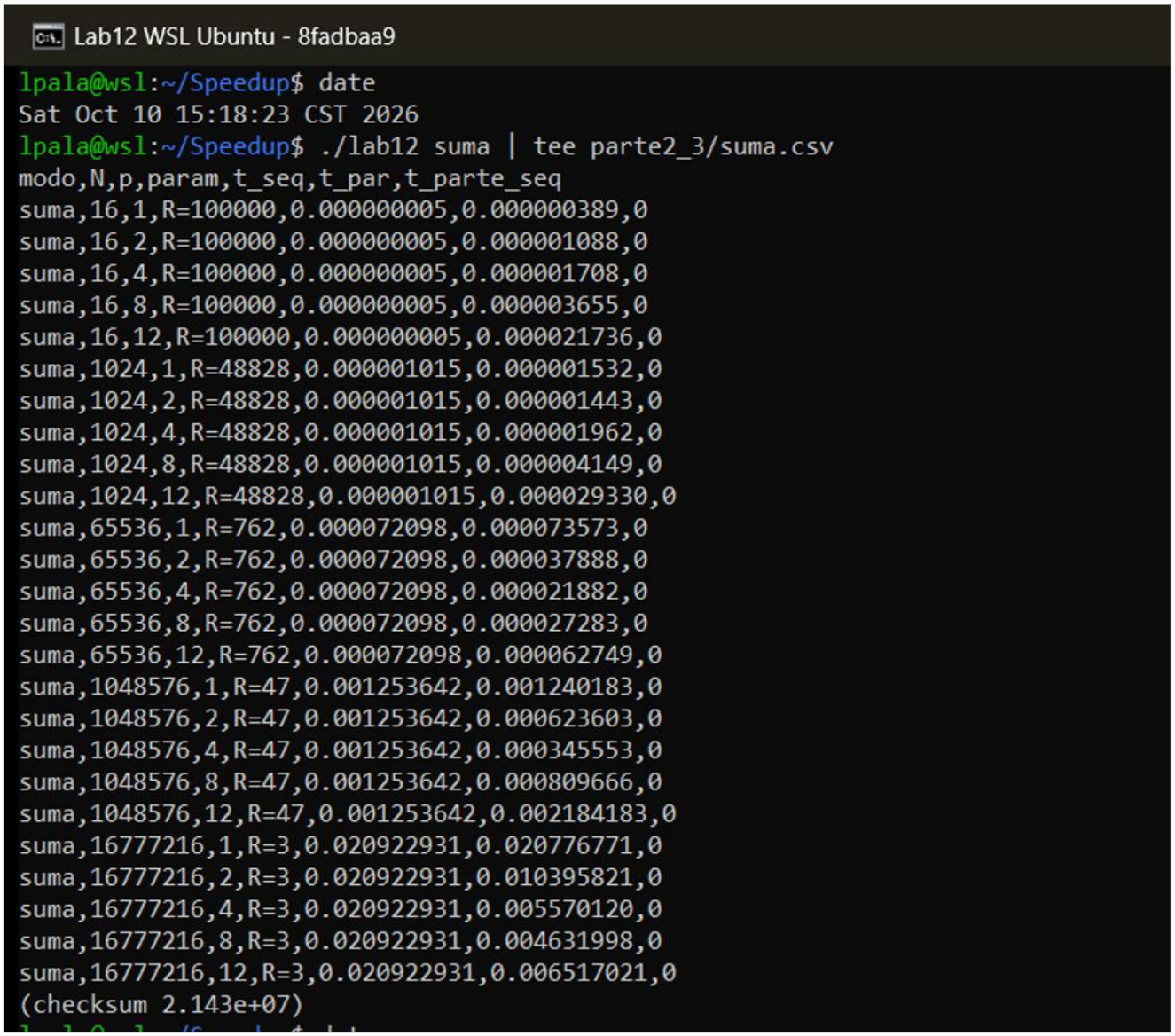
```
lpala@wsl:~/Speedup$ make clean
rm -f lab12 *.png *_con_S_E.csv
lpala@wsl:~/Speedup$ make
gcc -O2 -Wall -fopenmp -o lab12 lab12_speedup.c
lpala@wsl:~/Speedup$ ls -l lab12
-rwxrwxrwx 1 lpala lpala 21760 Oct 10  2026 lab12
lpala@wsl:~/Speedup$ ./lab12
Uso:
  ./lab12 amdahl     [W=400000] [fp=0.90] [pmax=12]
  ./lab12 suma       [pmax=12]
  ./lab12 desbalance [static|dynamic|guided] [W=40000] [pmax=12]
  ./lab12 gustafson  [W_por_hilo=100000] [W_seq=20000] [pmax=12]
```

Parte 2a: amdahl fp = 1.00 (para el Ts honesto)

Lab12 WSL Ubuntu - 823ba55e

```
lpala@wsl:~/Speedup$ date
Sat Oct 10 15:17:34 CST 2026
lpala@wsl:~/Speedup$ ./lab12 amdahl 400000 1.00 | tee parte2_3/amdahl_fp100.csv
modo,N,p,param,t_seq,t_par,t_parte_seq
amdahl,400000,1,fp=1.00,1.521067,1.478568,0.000000
amdahl,400000,2,fp=1.00,1.521067,0.745878,0.000000
amdahl,400000,3,fp=1.00,1.521067,0.474603,0.000000
amdahl,400000,4,fp=1.00,1.521067,0.368372,0.000000
amdahl,400000,5,fp=1.00,1.521067,0.297462,0.000000
amdahl,400000,6,fp=1.00,1.521067,0.253151,0.000000
amdahl,400000,7,fp=1.00,1.521067,0.223008,0.000000
amdahl,400000,8,fp=1.00,1.521067,0.234634,0.000000
amdahl,400000,9,fp=1.00,1.521067,0.186447,0.000000
amdahl,400000,10,fp=1.00,1.521067,0.189192,0.000000
amdahl,400000,11,fp=1.00,1.521067,0.177033,0.000000
(checksum 5.199e+12)
amdahl,400000,12,fp=1.00,1.521067,0.269922,0.000000
lpala@wsl:~/Speedup$ date
Sat Oct 10 15:18:04 CST 2026
```

Parte 2b: ./lab12 suma



Parte 3: ./lab12 desbalance static

Lab12 WSL Ubuntu - 42e0a045

```
lpala@wsl:~/Speedup$ date
Sat Oct 10 15:19:27 CST 2026
lpala@wsl:~/Speedup$ ./lab12 desbalance static | tee parte2_3/desb_static.csv
modo,N,p,param,t_seq,t_par,t_parte_seq
desbalance,40000,1,static,0.159815,0.146574,0
desbalance,40000,2,static,0.159815,0.109432,0
desbalance,40000,3,static,0.159815,0.083306,0
desbalance,40000,4,static,0.159815,0.063611,0
desbalance,40000,5,static,0.159815,0.052470,0
desbalance,40000,6,static,0.159815,0.046279,0
desbalance,40000,7,static,0.159815,0.040542,0
desbalance,40000,8,static,0.159815,0.034907,0
desbalance,40000,9,static,0.159815,0.040952,0
desbalance,40000,10,static,0.159815,0.032842,0
desbalance,40000,11,static,0.159815,0.028064,0
(checksum 5.199e+10)
desbalance,40000,12,static,0.159815,0.042762,0
lpala@wsl:~/Speedup$ date
Sat Oct 10 15:19:32 CST 2026
```

Parte 3: ./lab12 desbalance dynamic

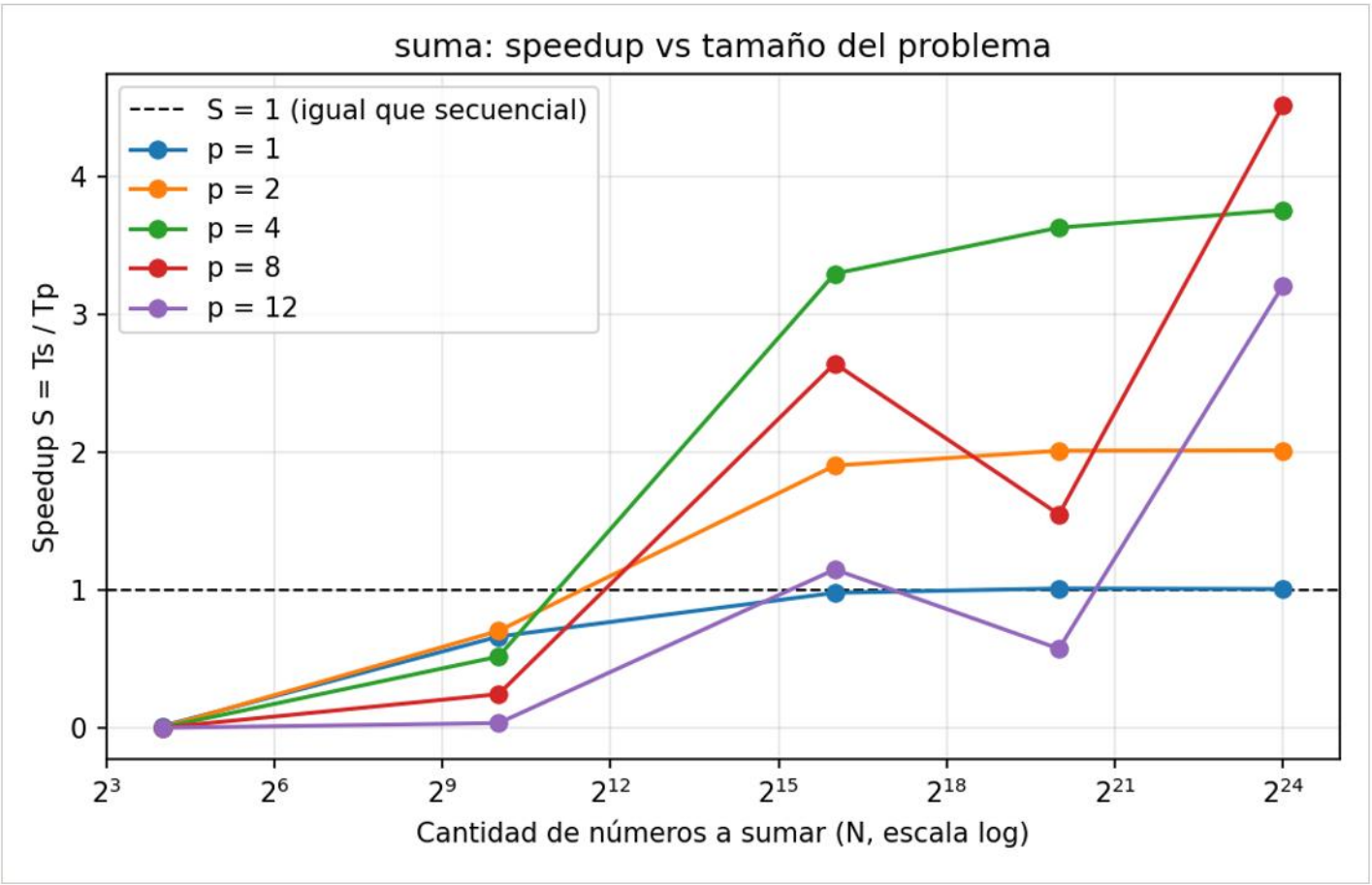
Lab12 WSL Ubuntu - 5df3781b

```
lpala@wsl:~/Speedup$ date
Sat Oct 10 15:19:35 CST 2026
lpala@wsl:~/Speedup$ ./lab12 desbalance dynamic | tee parte2_3/desb_dynamic.csv
modo,N,p,param,t_seq,t_par,t_parte_seq
desbalance,40000,1,dynamic,0.153325,0.153805,0
desbalance,40000,2,dynamic,0.153325,0.085414,0
desbalance,40000,3,dynamic,0.153325,0.052186,0
desbalance,40000,4,dynamic,0.153325,0.039029,0
desbalance,40000,5,dynamic,0.153325,0.031590,0
desbalance,40000,6,dynamic,0.153325,0.026222,0
desbalance,40000,7,dynamic,0.153325,0.020481,0
desbalance,40000,8,dynamic,0.153325,0.019168,0
desbalance,40000,9,dynamic,0.153325,0.018249,0
desbalance,40000,10,dynamic,0.153325,0.015767,0
desbalance,40000,11,dynamic,0.153325,0.024603,0
desbalance,40000,12,dynamic,0.153325,0.020480,0
(checksum 5.199e+10)
lpala@wsl:~/Speedup$ date
Sat Oct 10 15:19:38 CST 2026
```

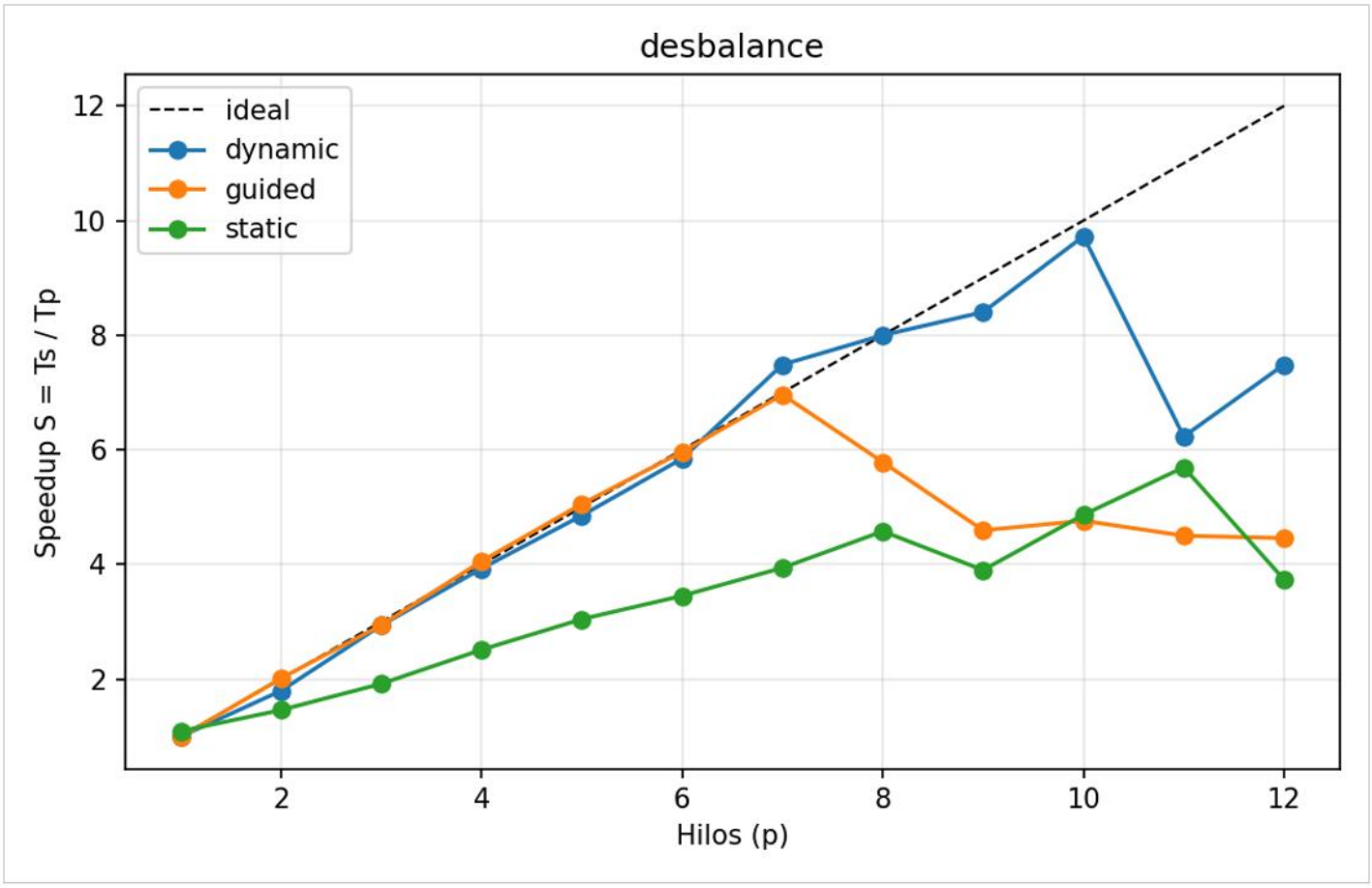
Parte 3: ./lab12 desbalance guided

```
Lab12 WSL Ubuntu - 70ddbcfc
lpala@wsl:~/Speedup$ date
Sat Oct 10 15:19:41 CST 2026
lpala@wsl:~/Speedup$ ./lab12 desbalance guided | tee parte2_3/desb_guided.csv
modo,N,p,param,t_seq,t_par,t_parte_seq
desbalance,40000,1,guided,0.153783,0.155471,0
desbalance,40000,2,guided,0.153783,0.076424,0
desbalance,40000,3,guided,0.153783,0.052269,0
desbalance,40000,4,guided,0.153783,0.037921,0
desbalance,40000,5,guided,0.153783,0.030418,0
desbalance,40000,6,guided,0.153783,0.025788,0
desbalance,40000,7,guided,0.153783,0.022085,0
desbalance,40000,8,guided,0.153783,0.026577,0
desbalance,40000,9,guided,0.153783,0.033463,0
desbalance,40000,10,guided,0.153783,0.032282,0
desbalance,40000,11,guided,0.153783,0.034162,0
desbalance,40000,12,guided,0.153783,0.034484,0
(checksum 5.199e+10)
lpala@wsl:~/Speedup$ date
Sat Oct 10 15:19:44 CST 2026
```

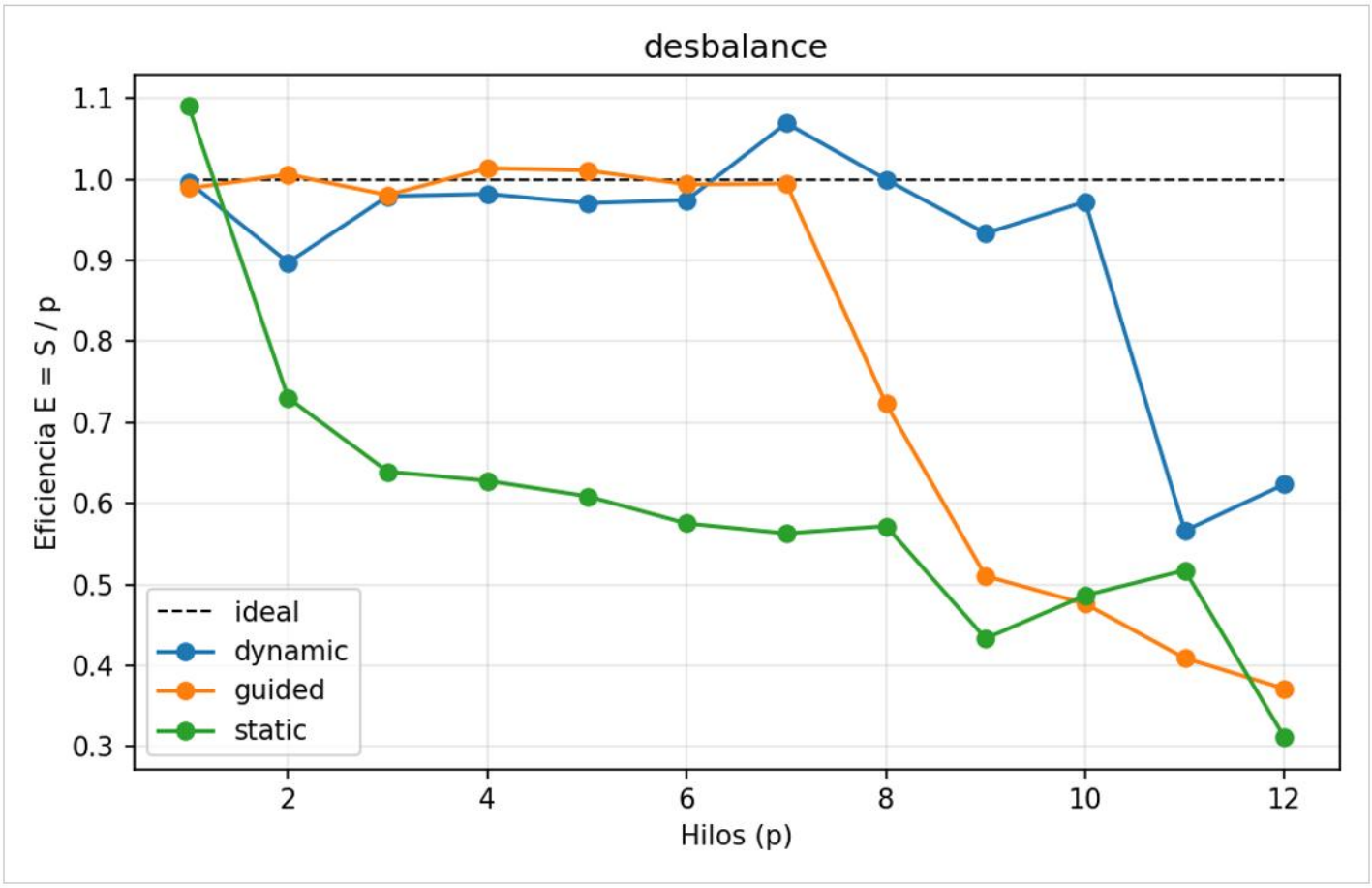
Gráfica: suma_speedup.png (graficar.py)



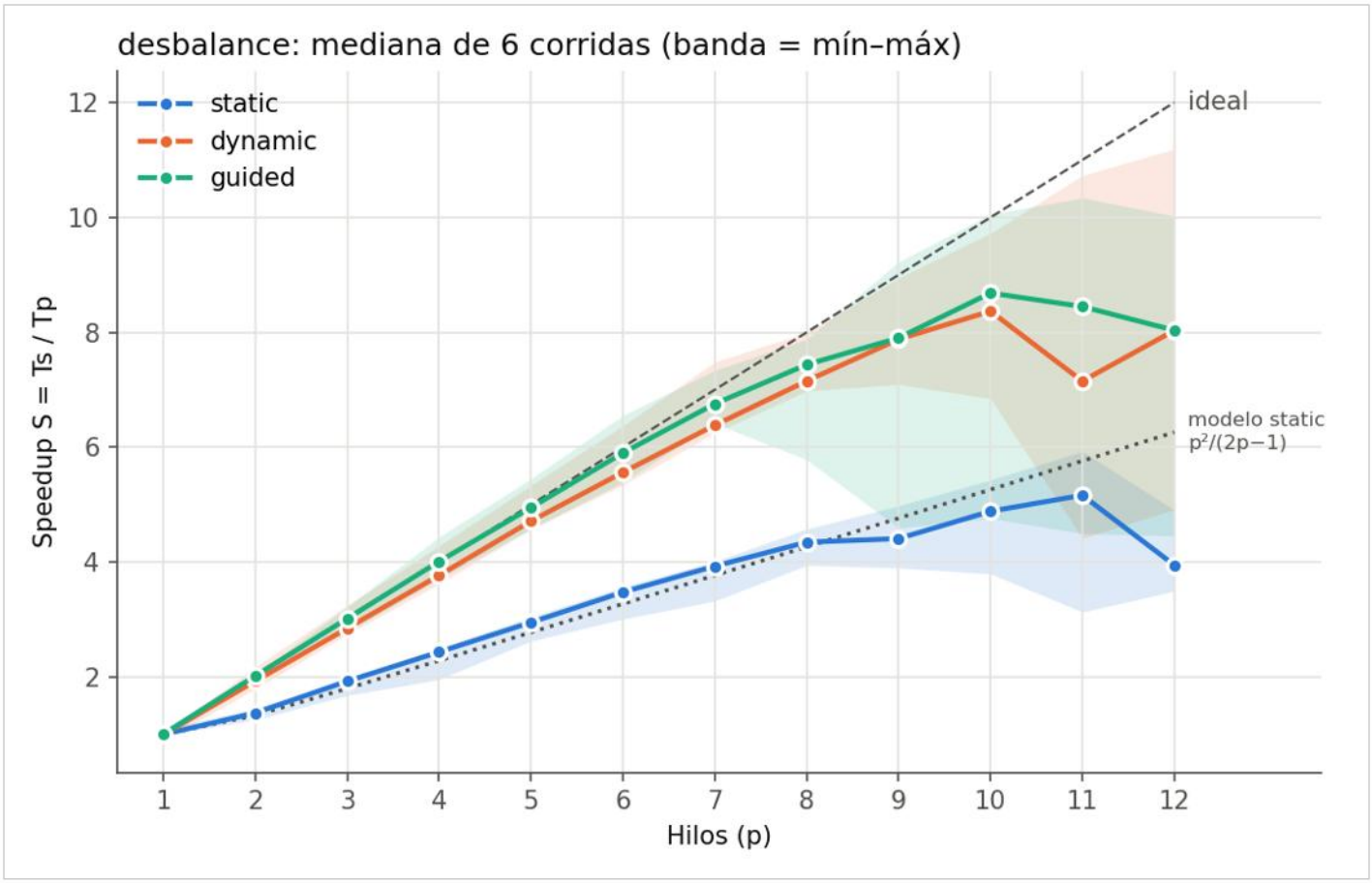
Gráfica: desbalance_speedup.png (graficar.py)



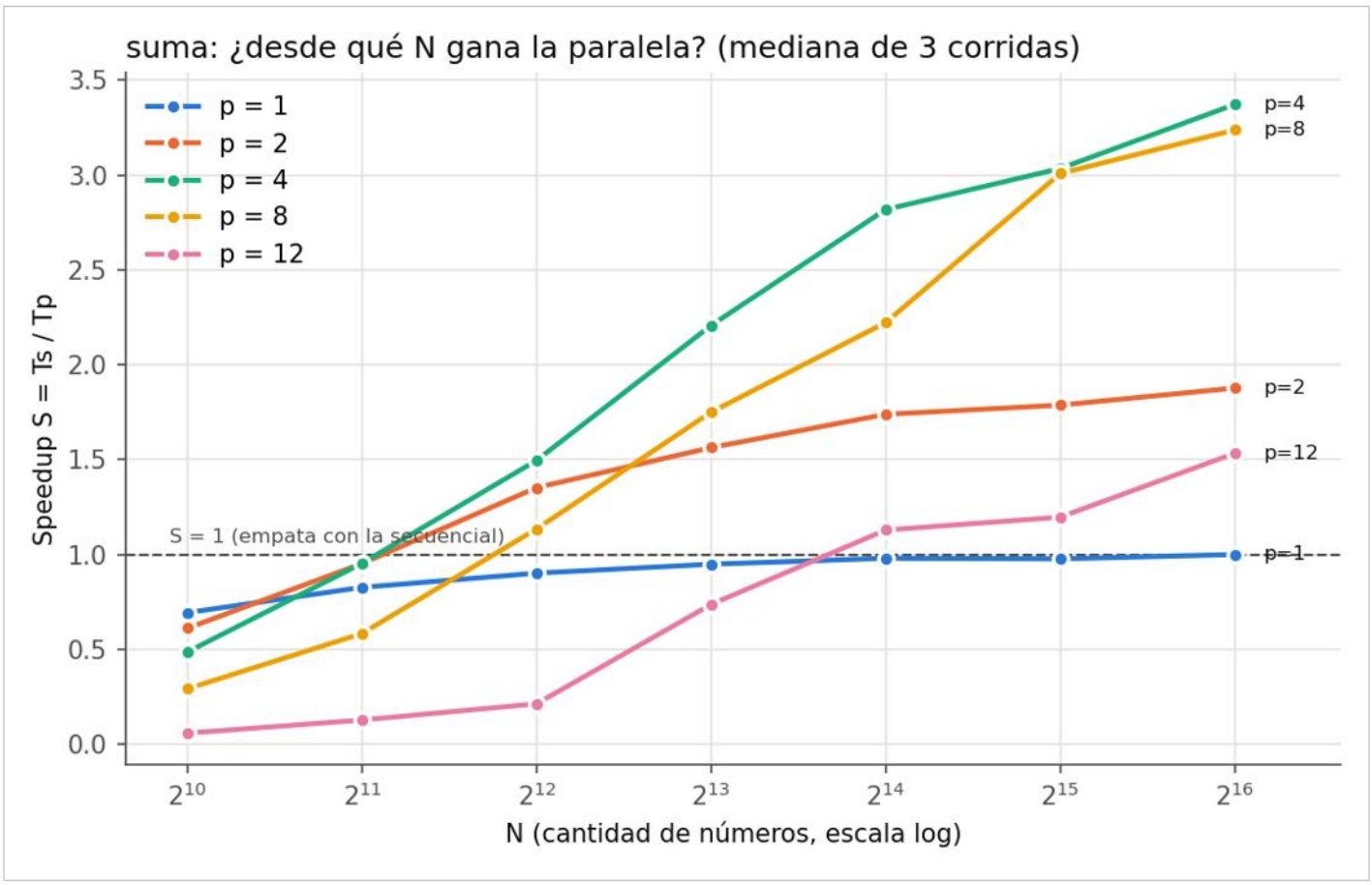
Gráfica: desbalance_eficiencia.png (graficar.py)



Extra: desbalance, mediana de 6 corridas



Extra: umbral de N para la suma (pregunta 2.6)



Reto: 5 corridas del lab12_speedup.c modificado

```
Lab12 WSL Ubuntu - 087eb37a
lpala@wsl:~/Speedup$ gcc -O2 -Wall -fopenmp -o lab12_reto lab12_speedup.c
corrida 1 (15:33:21):  N=262144 ->  p=2 S=2.34   p=4 S=4.44
corrida 2 (15:33:23):  N=262144 ->  p=2 S=3.61   p=4 S=5.80
corrida 3 (15:33:26):  N=262144 ->  p=2 S=2.57   p=4 S=6.74
corrida 4 (15:33:29):  N=262144 ->  p=2 S=2.28   p=4 S=6.22
corrida 5 (15:33:32):  N=262144 ->  p=2 S=2.80   p=4 S=5.03
```

Reto: eficiencia > 1 (S > p)

